Bags, Vectors & Transformers

Themes and Span Coding · Bonus, Part 16

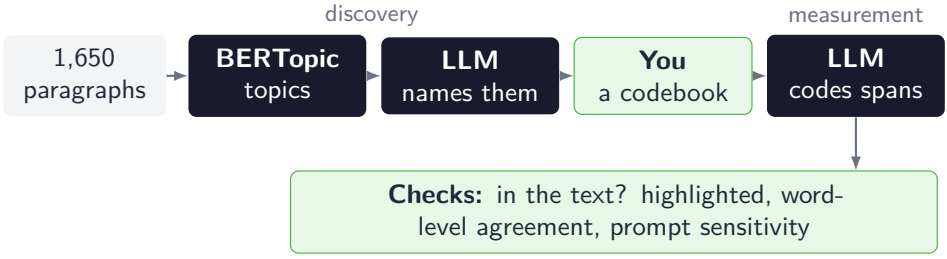
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Discovery, Then Measurement

	Topic model (BERTopic)	**Codebook + span coding** (LLM)
Question	which themes are in these texts?	where exactly is theme X, and how much?
Themes defined by	the data	you
Unit	a document, one topic	any passage; several themes per paragraph
Strength	finds what you didn't expect	measures what you mean, checkably

Used together they follow a classic route: **explore, define, measure, validate** (Nelson, 2020). Bonus 11 covers BERTopic in depth; here it is only the first step.

The Pipeline



Data: the 60 US inaugural addresses in NLTK (1789–2025), as in Bonus 10, split into paragraphs of at most 160 words. The LLM is Qwen2.5 7B in Ollama, with structured outputs as in Bonus 15.

Names Are Hypotheses

For each large topic the LLM sees the **top words** and three **representative paragraphs**, and returns a name and a one-sentence definition.

A name can be fluent and still describe the top words rather than the paragraphs. Read the paragraphs behind every topic before you trust a label, from a person or a model.

From topics to a codebook

- merge topics that are one theme
- drop "topics" held together by ceremony or style
- keep themes broad enough to recur across eras
- write definitions a coder can apply

The notebook starts with five themes: economy, war and peace, liberty, unity, faith. Edit them.

What Is Span Coding?

Not one label per paragraph, but the **exact passages** for each theme. One possible coding of Lincoln, 1865:

> With malice toward none, with charity for all, **with firmness in the right as God gives us to see the right**, let us strive on to finish the work we are in, **to bind up the nation's wounds**, **to care for him who shall have borne the battle and for his widow and his orphan**, **to do all which may achieve and cherish a just and lasting peace among ourselves and with all nations**.

God and faith **Unity and national purpose** **War, peace and security** (the notebook highlights them in color)

The model returns a list of `{theme, quote}`; the schema allows only codebook themes.

Spans make the coding **inspectable**: you can see, check and compare exactly what was coded.

Checking Spans

In the text?
Locate every span: exact, partly ($\geq 60\%$), or not in text. The positions drive everything else.

Highlighted
Colored passages show at a glance what was coded, and what wasn't: the fastest check there is.

Word by word
Agreement between two codings: the share of coded words both marked (F1), per theme.

Prompt sensitivity. Coding the same paragraphs with the themes listed in reverse order should change nothing, and often does. Fix the prompt and the codebook before the final run.

For human coders, the formal measure is Krippendorff's unitizing alpha.

Rules for Spans, and Themes over Time

Rules the codebook needs

- span length: “the shortest clause that expresses the theme”
- can spans overlap? can a paragraph have none?
- every new theme takes spans from its neighbors: freeze the codebook, then validate

Emphasis over time

The **share of words** coded for each theme, per era or per speech: Roosevelt 1933 next to Reagan 1981.

60 paragraphs give hypotheses. All 1,650 take about two hours on a T4, after validating on a hand-coded sample.

In the Notebook

- 1,650 paragraphs from 60 inaugural addresses
- BERTopic topics, named by an LLM, read by you
- From topics to a codebook you control
- Span coding with structured outputs, checked and highlighted
- Word-level agreement: prompt sensitivity, and your own coding
- Spans versus topics, and themes across two centuries

`bonus/notebooks/16_themes_and_span_coding_exercises.ipynb` · 5 exercises · GPU